

SOCIOL 723: Problem Set 2

Assigned: Friday, September 18 • **Due:** Friday, October 2, 11:59PM

Your name:

Instructions. Submit a single PDF rendered from a Quarto or RMarkdown source, together with the source file. This problem set is about what you will do in research: reading a published table, fitting the same kind of model on real data, and writing up what it says. No question asks for a derivation, and every computation uses the functions from Labs 3 and 4 (`glm`, `marginalEffects`, `sandwich`, `AER`, `MASS`, `pscl`, `lme4`, `nlme`). Where a question asks for sentences, write them as you would in a paper. You may use AI assistance, but you are responsible for understanding and being able to explain every line you submit. Points per item are in brackets.

Use `set.seed(723)` wherever there is randomness. The GSS extract is `Data/gss_earnings.rds`; the school data are `nlme::MathAchieve` and `nlme::MathAchSchool`, merged as in Lab 4.

Reading a Published Logit: Killewald (2016) (25 points)

Killewald, Alexandra. 2016. “Money, Work, and Marital Stability: Assessing Change in the Gendered Determinants of Divorce.” *American Sociological Review* 81(4): 696–719. Killewald follows couples in the Panel Study of Income Dynamics year by year. One row of her data is one couple in one year, and the outcome is whether the couple divorced in that year. Table 3 fits a logit separately to marriages begun in 1974 or earlier and to marriages begun in 1975 or later. The excerpt below keeps the rows the questions use; the full table has about forty rows of controls.

	Married 1974 or earlier			Married 1975 or later		
	$\hat{\beta}$	(SE)	OR	$\hat{\beta}$	(SE)	OR
Wife employed full-time	.267	(.153)	1.306	.069	(.088)	1.071
Husband employed full-time	−.097	(.208)	.908	−.293**	(.106)	.746
Wife’s proportion of housework	−1.117*	(.463)	.327	.050	(.233)	1.051
Constant	−7.476***	(.884)		−3.525***	(.393)	
Couple-years	32,853			33,470		

Killewald (2016), Table 3, excerpt. Logit coefficients, standard errors, and odds ratios as printed. * $p < .05$; ** $p < .01$; *** $p < .001$. Controls include marital duration, religion, race, home ownership, children, both spouses’ education, region, and year of marriage.

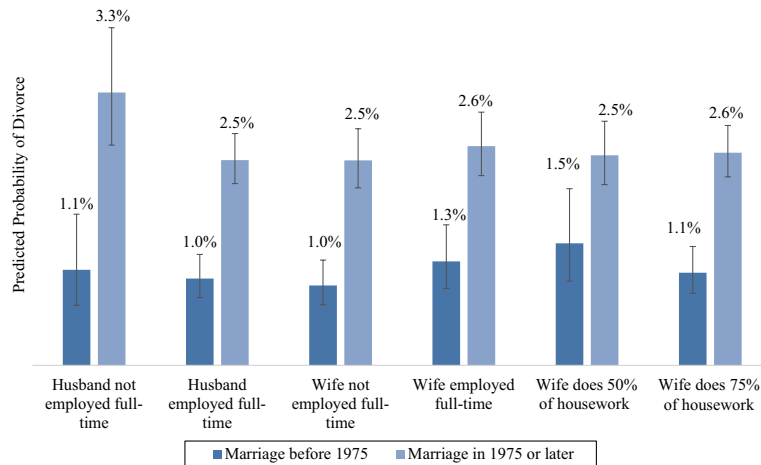


Figure 1. Predicted Probabilities of Divorce in the Next Year
Note: Probabilities are predicted based on the logistic regression coefficients in Table 3 for a respondent with all other covariates set to the cohort-specific means. Error bars are 95% confidence intervals, estimated by calculating the bounds of the confidence interval for the predicted log odds (non-bootstrapped) and then transforming those bounds to predicted probabilities.

Figure 1: Killewald (2016), Figure 1, reproduced for classroom use.

1. **The data.** [5] In two or three sentences: what does the outcome equal in a given row, why does a couple that stays married for twenty years contribute twenty rows, and what would be lost by collapsing the data to one row per couple with the outcome “ever divorced”?
2. **One coefficient, three scales.** [7] Take the husband’s full-time employment in the later cohort, $-.293$. Write one sentence on each scale: log-odds, odds (use the printed odds ratio), and probability (from Figure 1: 2.5% when he is employed full-time, 3.3% when he is not). Then compute the odds ratio implied by those two probabilities and compare it with the printed .746. Should the two agree exactly? Answer in one or two sentences, using the note under Figure 1 on how the probabilities were computed.
3. **A coefficient that is not significant.** [7] In the later cohort, the wife’s full-time employment has coefficient .069 and standard error .088; Figure 1 shows 2.6% when she is employed full-time and 2.5% when she is not. Write the sentence that belongs in a results section. In the earlier cohort the same coefficient is .267 with standard error .153, and Figure 1 shows 1.3% against 1.0%. Killewald writes that the association declined across cohorts but that the decline is not statistically significant. Compute the difference between the two coefficients and its standard error, $\sqrt{.153^2 + .088^2}$, and use them to explain in two sentences what supports each half of her claim.
4. **Units matter.** [6] The wife’s proportion of housework runs from 0 to 1, and its coefficient in the earlier cohort is -1.117 , printed odds ratio .327. What change in housework does that odds ratio describe? Figure 1 plots a smaller change, from 50% to 75%. Start from the figure’s 1.5% at 50%, multiply the odds by $e^{-1.117 \times .25}$, and convert back to a probability; you should land on the figure’s 1.1%. Then one sentence on why you cannot recover

the odds ratio precisely by working backwards from the printed 1.5% and 1.1%, and one sentence on why a coefficient's size means nothing until you know the units of its variable.

Your Own Logit, Reported Like a Paper (30 points)

Fit the logit from lecture on the GSS extract:

```
m1 <- glm(ba ~ pareduc + wordsum + female + black, data = gss, family = binomial)
```

1. **Table and figure.** [10] Report the coefficient table with odds ratios added, laid out like Killewald's Table 3. Then make your own Figure 1: with `avg_predictions`, compute the average predicted probability of a degree when everyone's parents have 8, 12, 16, and 20 years of education, with 95% intervals, and plot the four bars with error bars. Two sentences: why is the gain from 12 to 16 years not the same as the gain from 16 to 20, when the coefficient is one number?
2. **An interaction, read correctly.** [8] Does parents' education matter more for women? Fit `ba ~ pareduc * female + wordsum + black`. Report the product coefficient with its z , then the AME of `pareduc` separately for men and women and the difference with its standard error. Answer the question in a short paragraph and say why the product coefficient alone could not have answered it.
3. **A block of controls.** [6] Add the region dummies (`region`) to `m1`. Report the likelihood-ratio test, AIC, and BIC for the two models. In a short paragraph, say what each is measuring, why with $n = 3,509$ the LR test rejects almost any block, and which model you would report and why.
4. **The results paragraph.** [6] Write the paragraph (120 to 180 words) that would appear in the Results section of a paper reporting this model. Use predicted probabilities or AMEs, not odds ratios; give the reader magnitudes and uncertainty; do not mention the log-odds scale.

Counts (20 points)

Fit the Poisson regression for the number of children:

```
m_pois <- glm(childs ~ educ + female + black + exper + evermar, data = gss, family = poisson)
```

1. **Reading it, and which standard errors.** [8] Interpret $e^{\hat{\beta}}$ for `educ` and for `evermar`, one sentence each, in the form you would use in a paper. Then run `AER::dispersiontest(m_pois)` and report the standard error of `evermar` from the Poisson fit, from a quasi-Poisson fit, and from `sandwich::vcovHC`. Two sentences: what does the test say about the Poisson's variance assumption, and which standard error would you report as your default and why?

2. **Zeros, and a model with two parts.** [12] Report the observed share of childless respondents and the share the Poisson predicts (average `dpois(0, fitted(m_pois))`). Fit the hurdle model with `pscl::hurdle(..., dist = "poisson")` and the same covariates in both parts. Make one table with the log-likelihood, number of parameters, AIC, and BIC of the Poisson and the hurdle. Then, in a paragraph written for a reader, say what **evermar** does to the chance of *any* child, what it does to the number of children *among parents*, and what the single Poisson coefficient was averaging over.

Observations in Groups (25 points)

Merge `nlme::MathAchieve` with `nlme::MathAchSchool` as in Lab 4, and convert `School` to a plain factor with `factor(as.character(School))`, as the lab does: 7,185 students in 160 schools, math achievement, student SES, and school sector.

1. **From clustered standard errors to a random intercept.** [13] Fit `lm(MathAch ~ SES + Sector)` and report the coefficient on `Sector` with its plain standard error and with its standard error clustered on `School`. In a short paragraph, explain to a first-year student why the plain standard error is wrong for this variable: sector is a property of the school, so it varies across 160 schools, not across 7,185 students. Then fit `lmer(MathAch ~ SES + Sector + (1 | School), REML = FALSE)` and report the two coefficients with standard errors, $\hat{\tau}^2$, and $\hat{\sigma}^2$. Say in one sentence each what the two variances measure, compute the intraclass correlation, and close with two sentences on what the random-intercept model gives you that the clustered standard error did not.
2. **Three estimates of one slope, and three questions.** [12] Fit the fixed-effects regression `lm(MathAch ~ School + SES + Sector)`, with the school dummies listed *before* `Sector`. One coefficient comes back NA: which one, and why? What would you have concluded had you listed `Sector` before `School` and read off the number R printed? Then compare the SES slope under pooled OLS, the random intercept, and fixed effects in one three-column table, and explain in a paragraph where the random-intercept slope sits relative to the other two and why, in terms of the variation each estimate uses. Finally, for each research question below, say in two sentences which of *OLS with clustered standard errors*, *fixed effects*, or a *random intercept* you would use, naming the feature of the question that decides it.
 - (a) How much of earnings inequality in a country is between workplaces rather than within them, and is that share rising?
 - (b) Does a worker's wage rise after their own firm signs a union contract, using several years of data on the same workers and firms?
 - (c) Do students learn more in Catholic schools, using these 160 schools, where sector is a property of the school?

Total: 100 points.